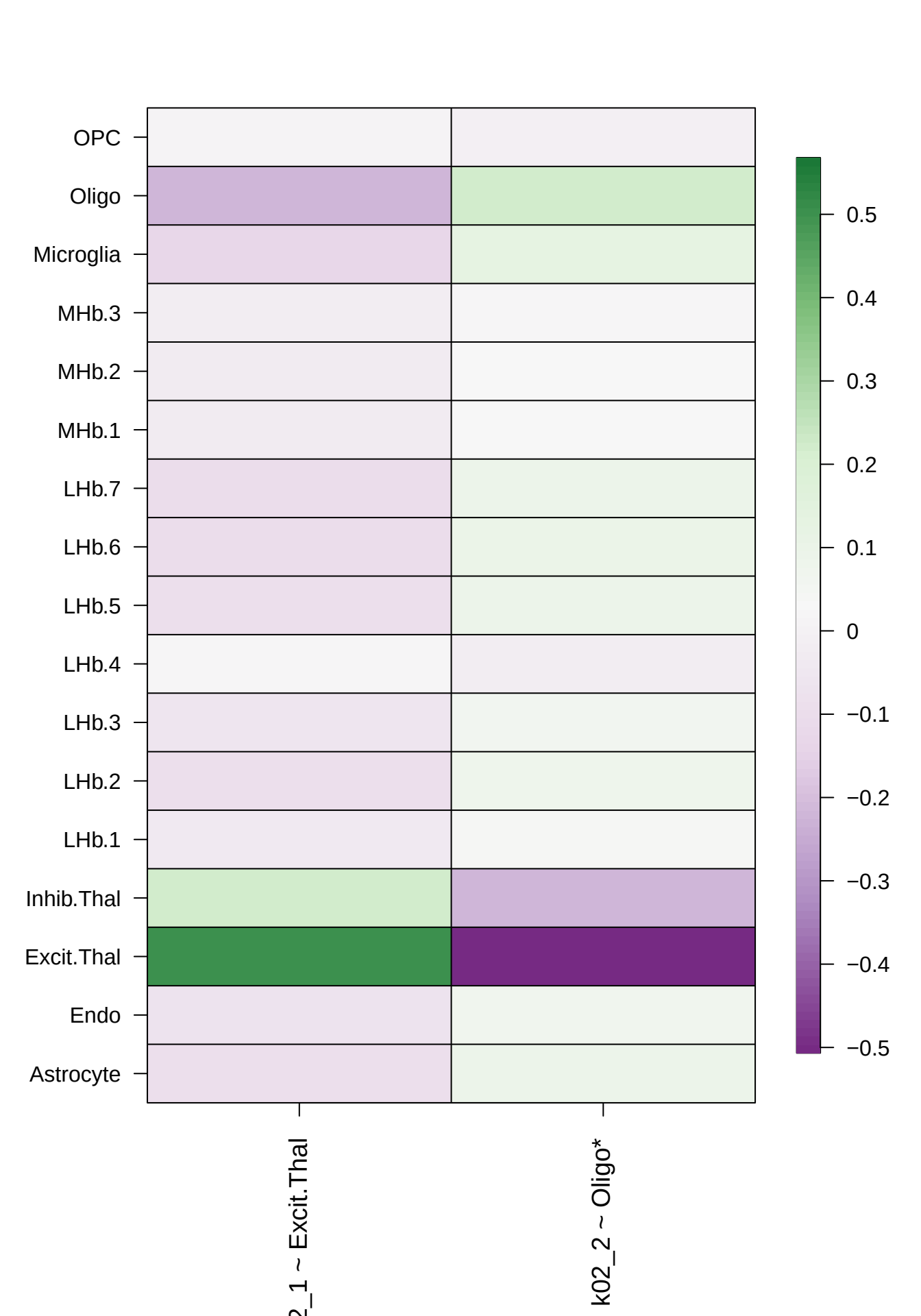
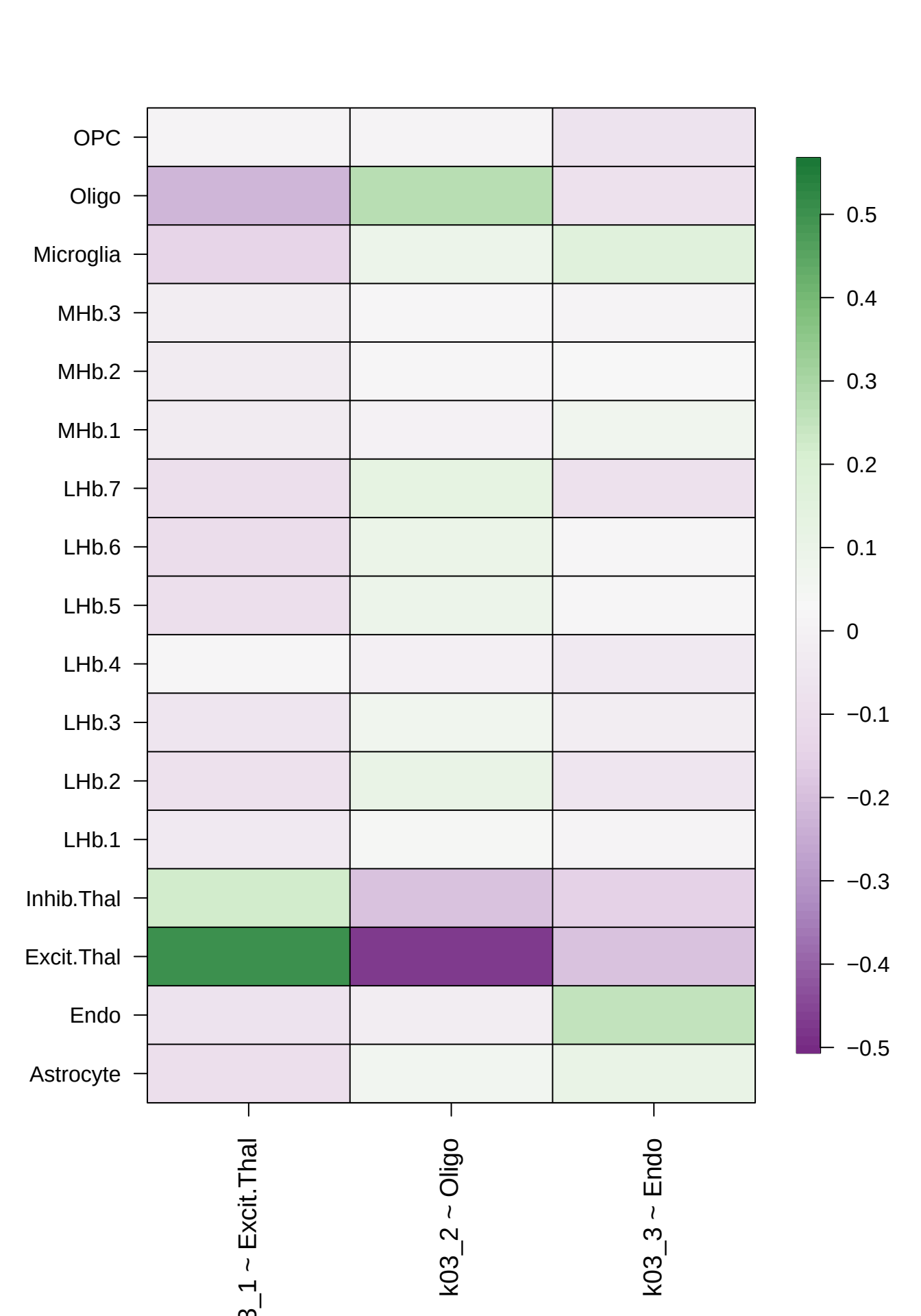
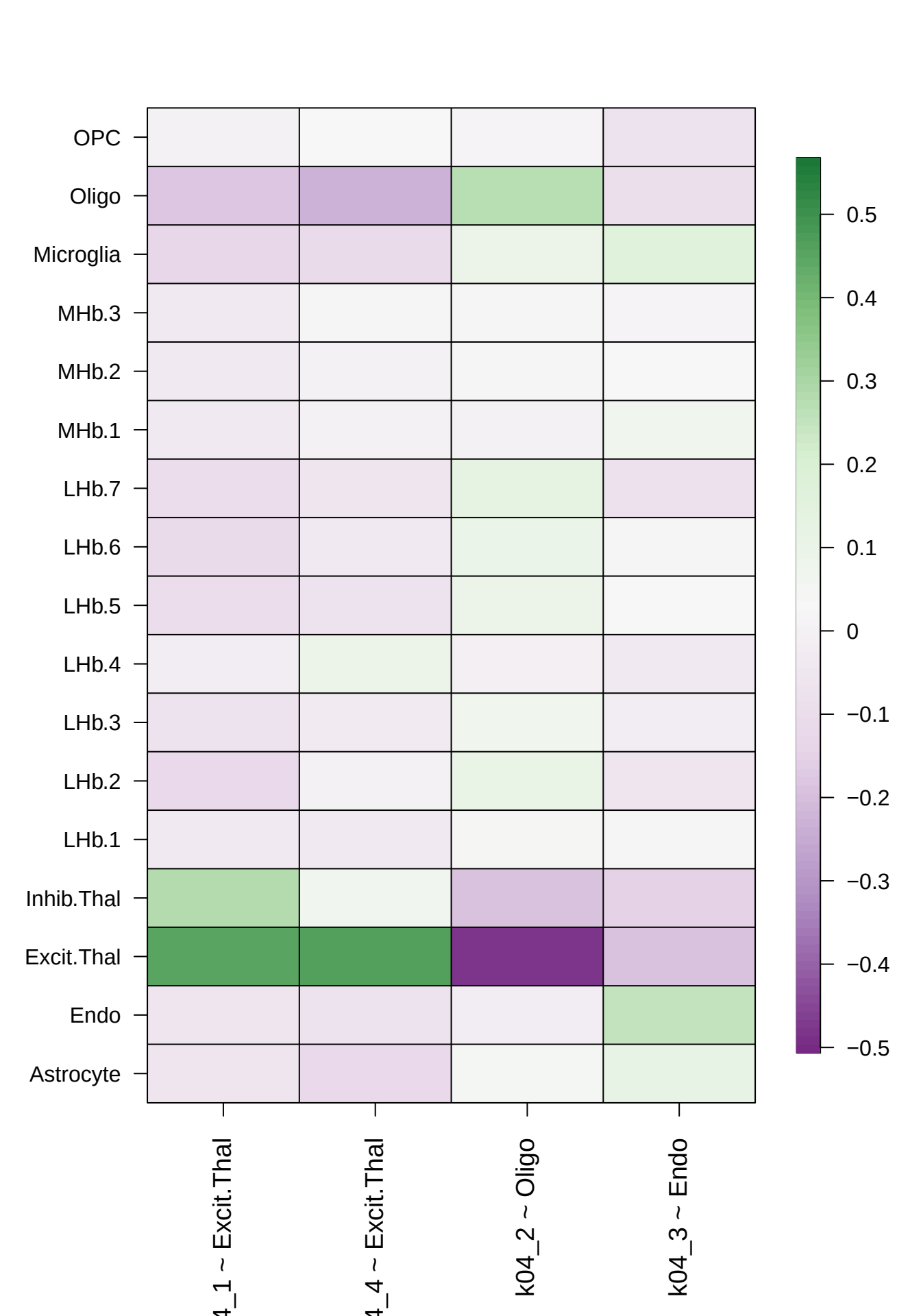
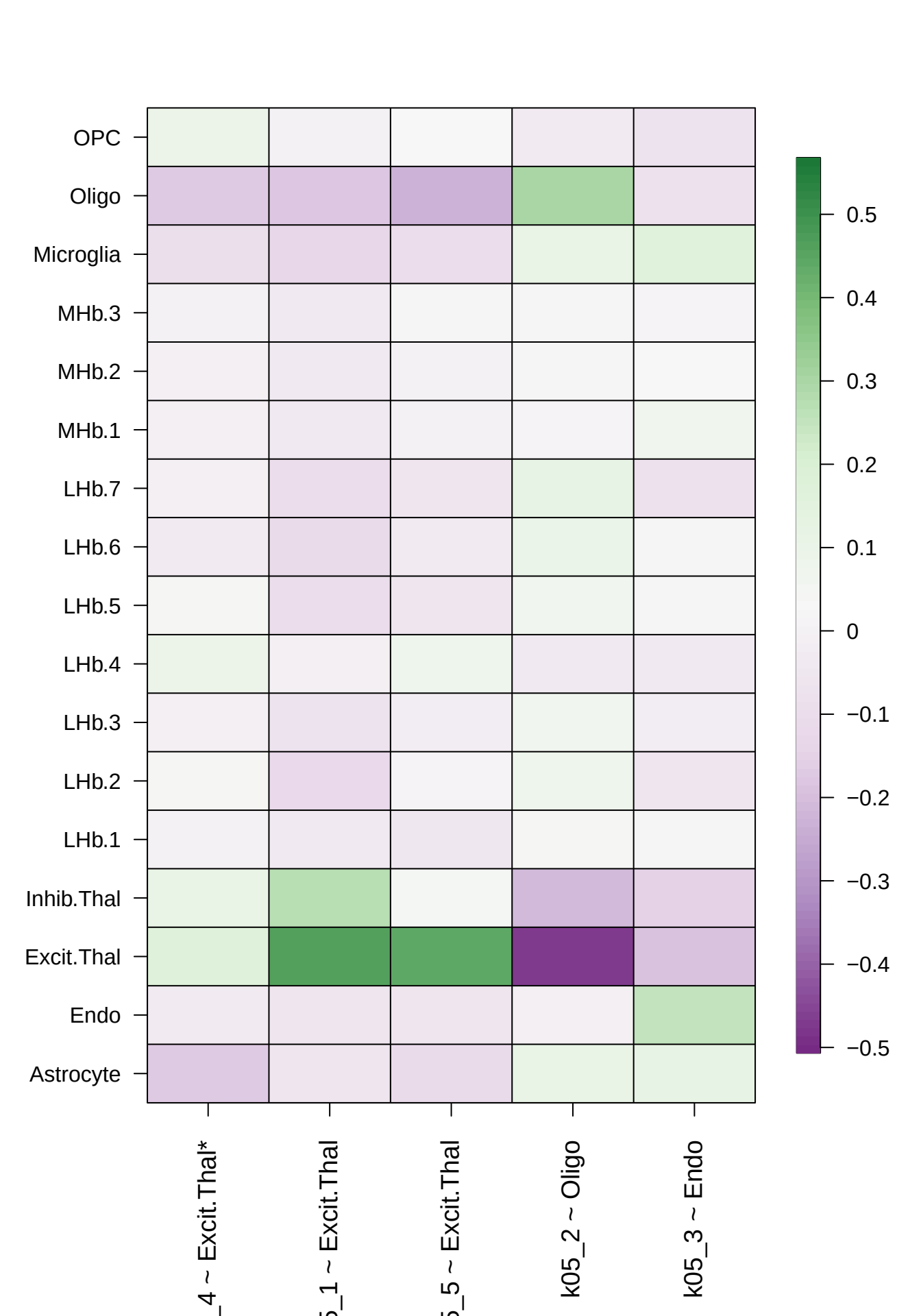
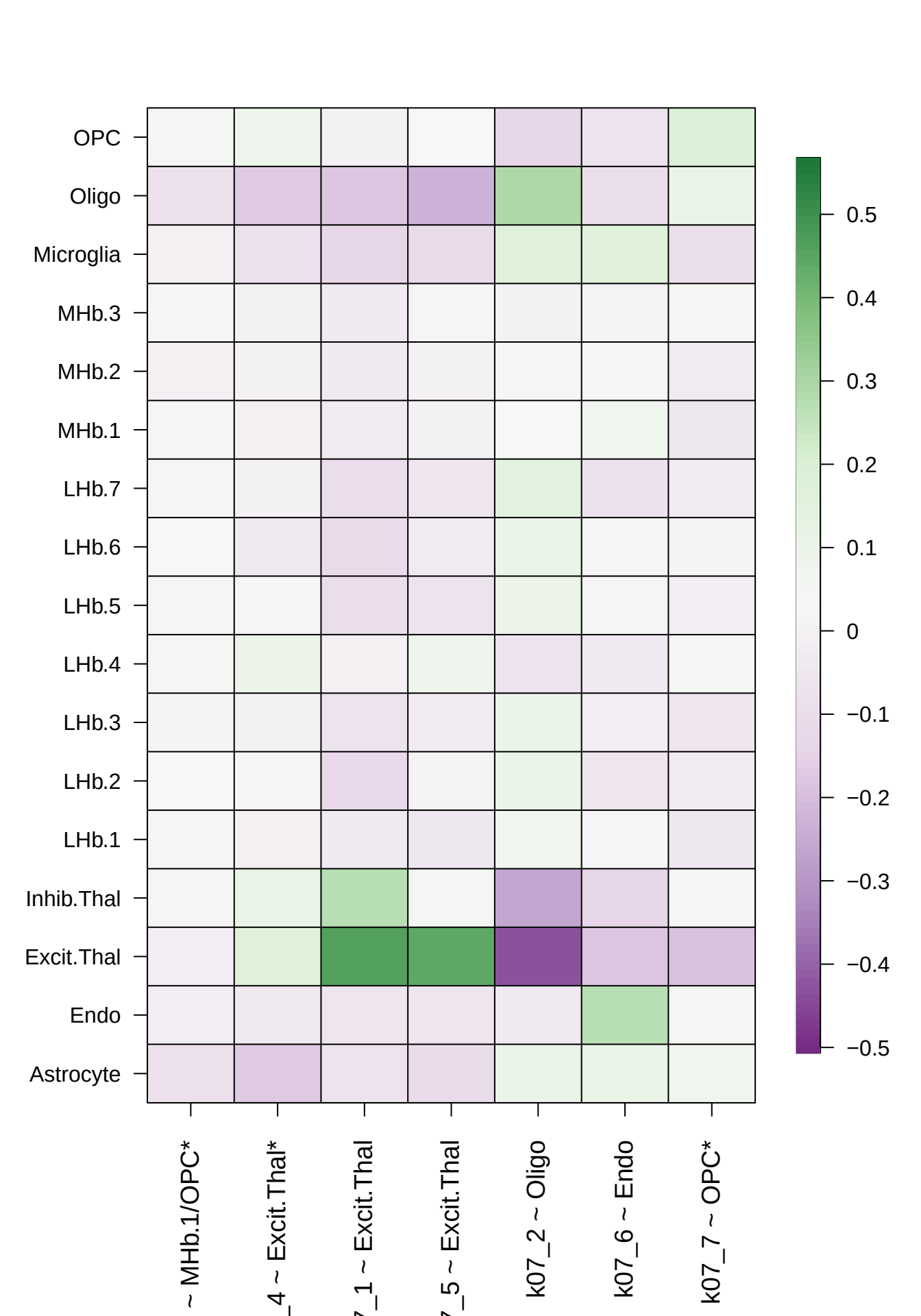

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
k02_2 ~ Oligo*

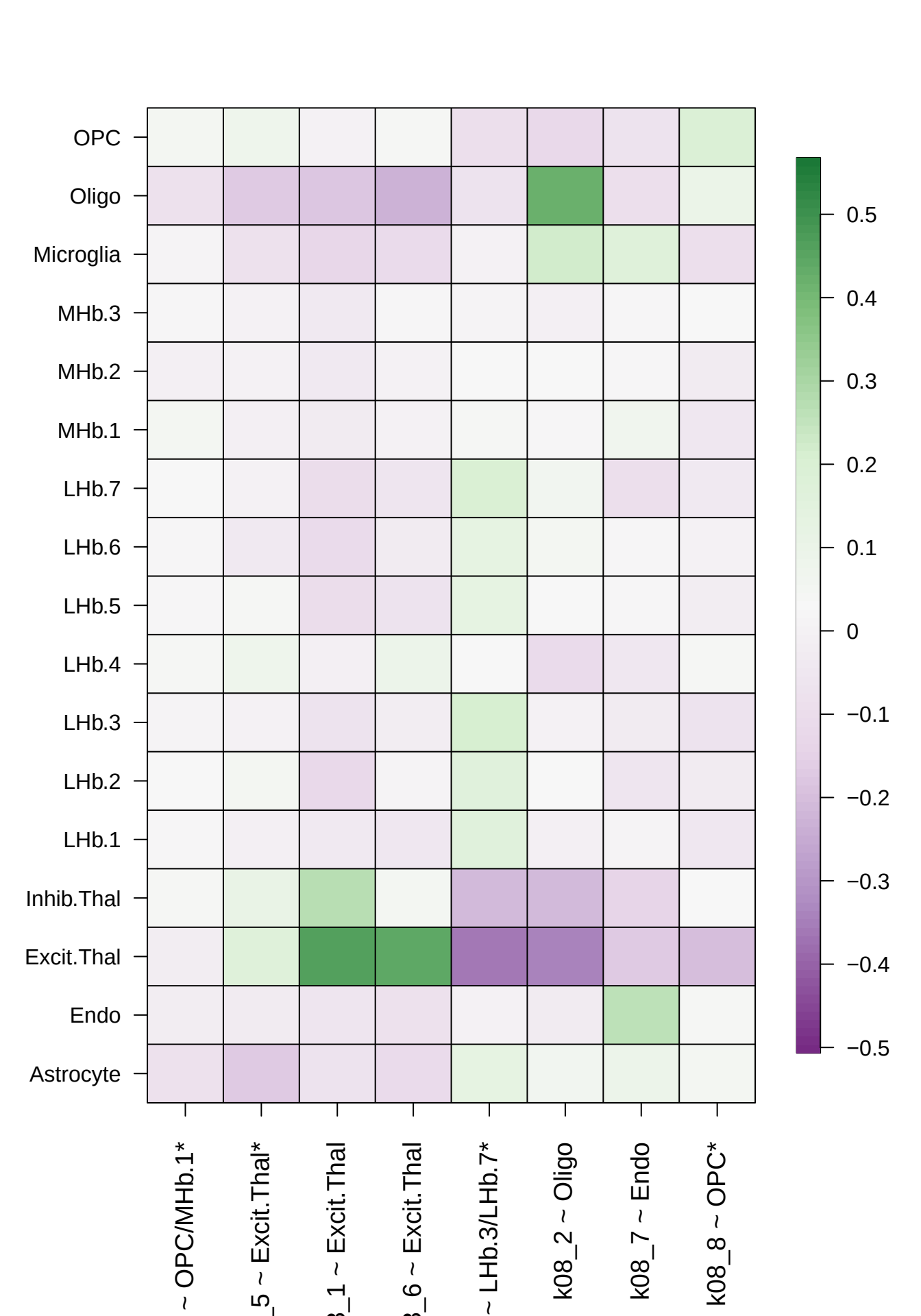

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
k03_1 ~ Excit.Thal
k03_2 ~ Oligo
k03_3 ~ Endo
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

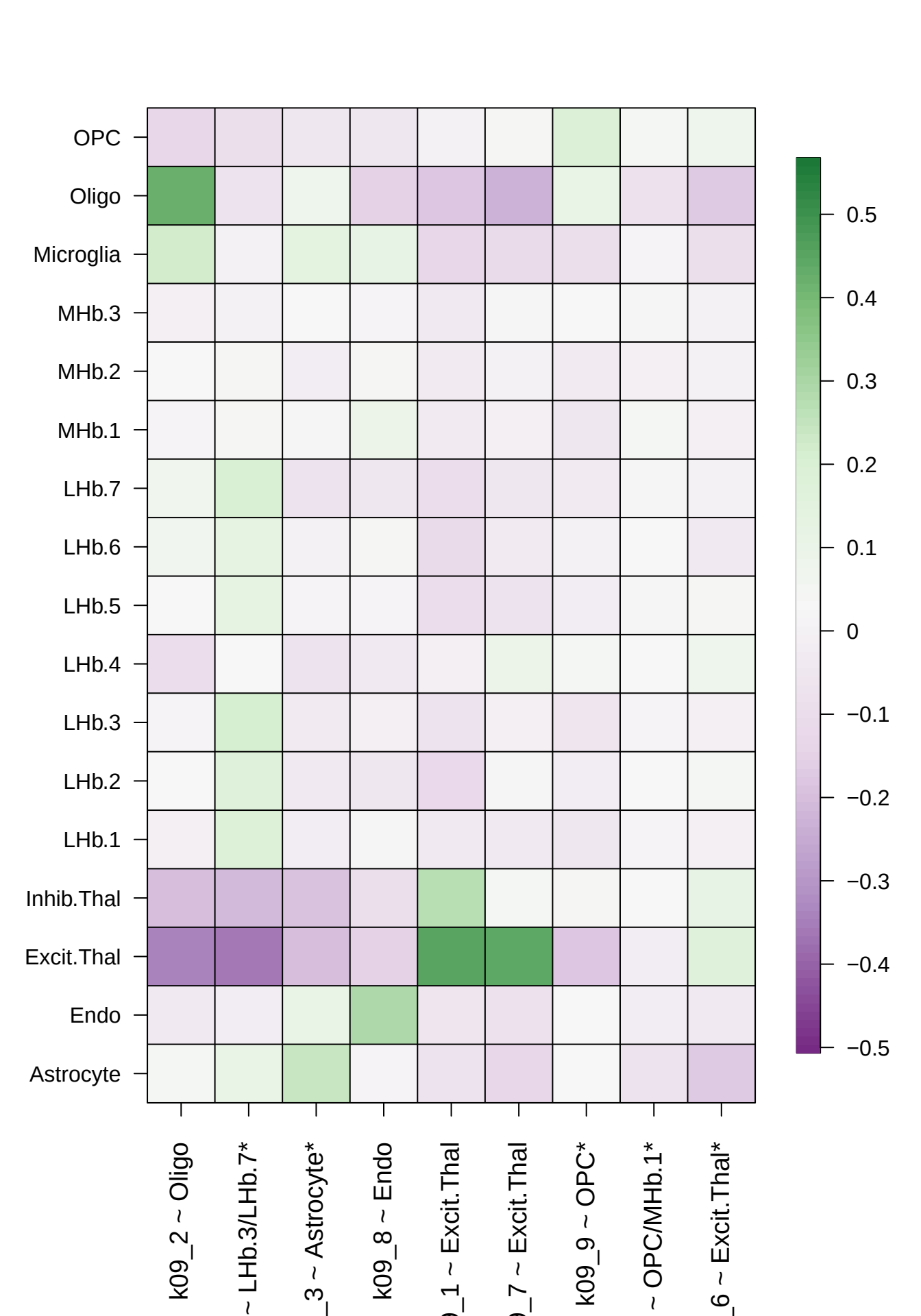

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
k04_1 ~ Excit.Thal
k04_4 ~ Excit.Thal
k04_2 ~ Oligo
k04_3 ~ Endo
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

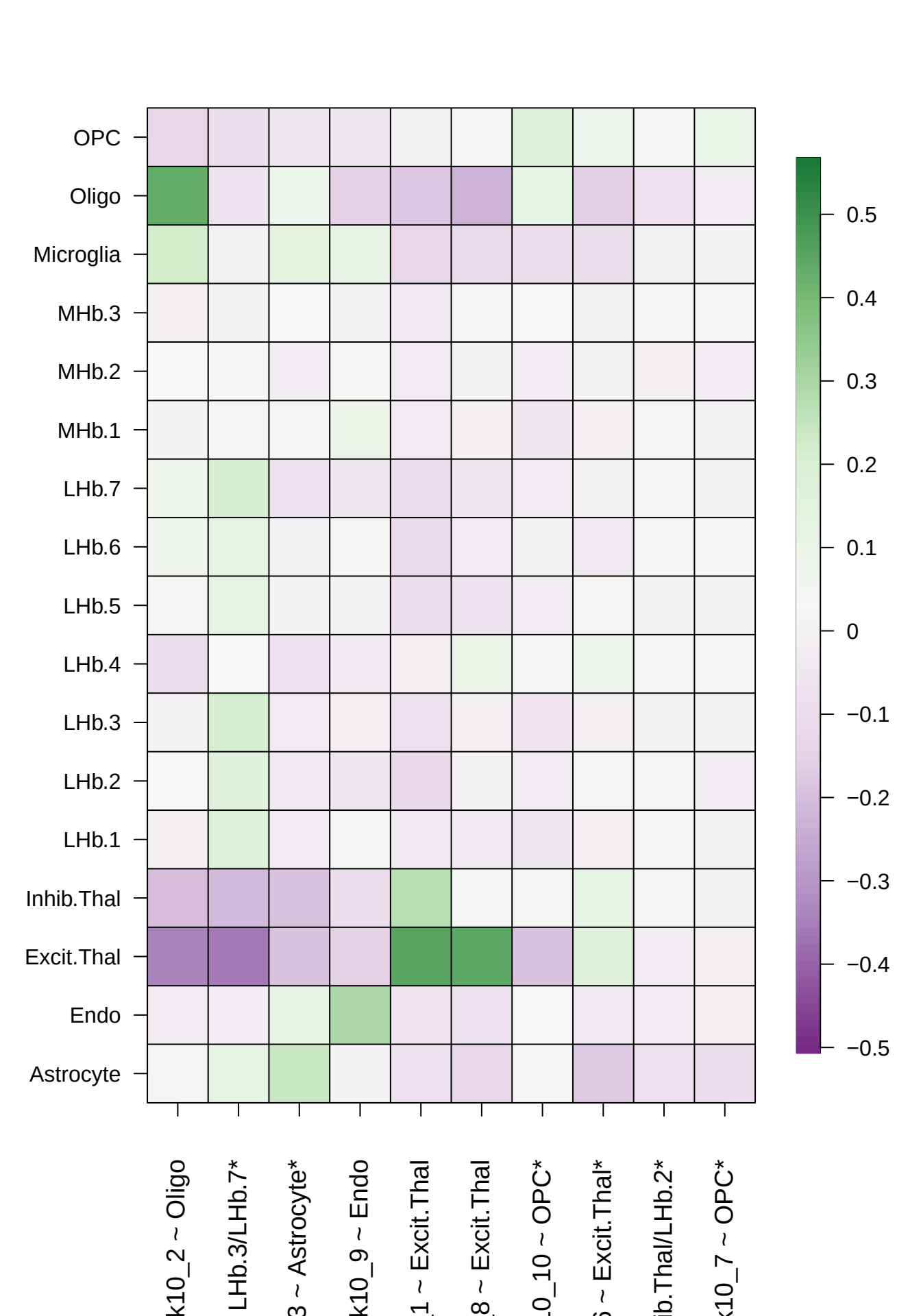

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
_4 ~ Excit.Thal*
_1 ~ Excit.Thal
_5 ~ Excit.Thal
k05_2 ~ Oligo
k05_3 ~ Endo
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
k06_4 ~ Excit.Thal*
k06_1 ~ Excit.Thal
k06_5 ~ Excit.Thal
k06_2 ~ Oligo
k06_3 ~ OPC*
k06_6 ~ Endo
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

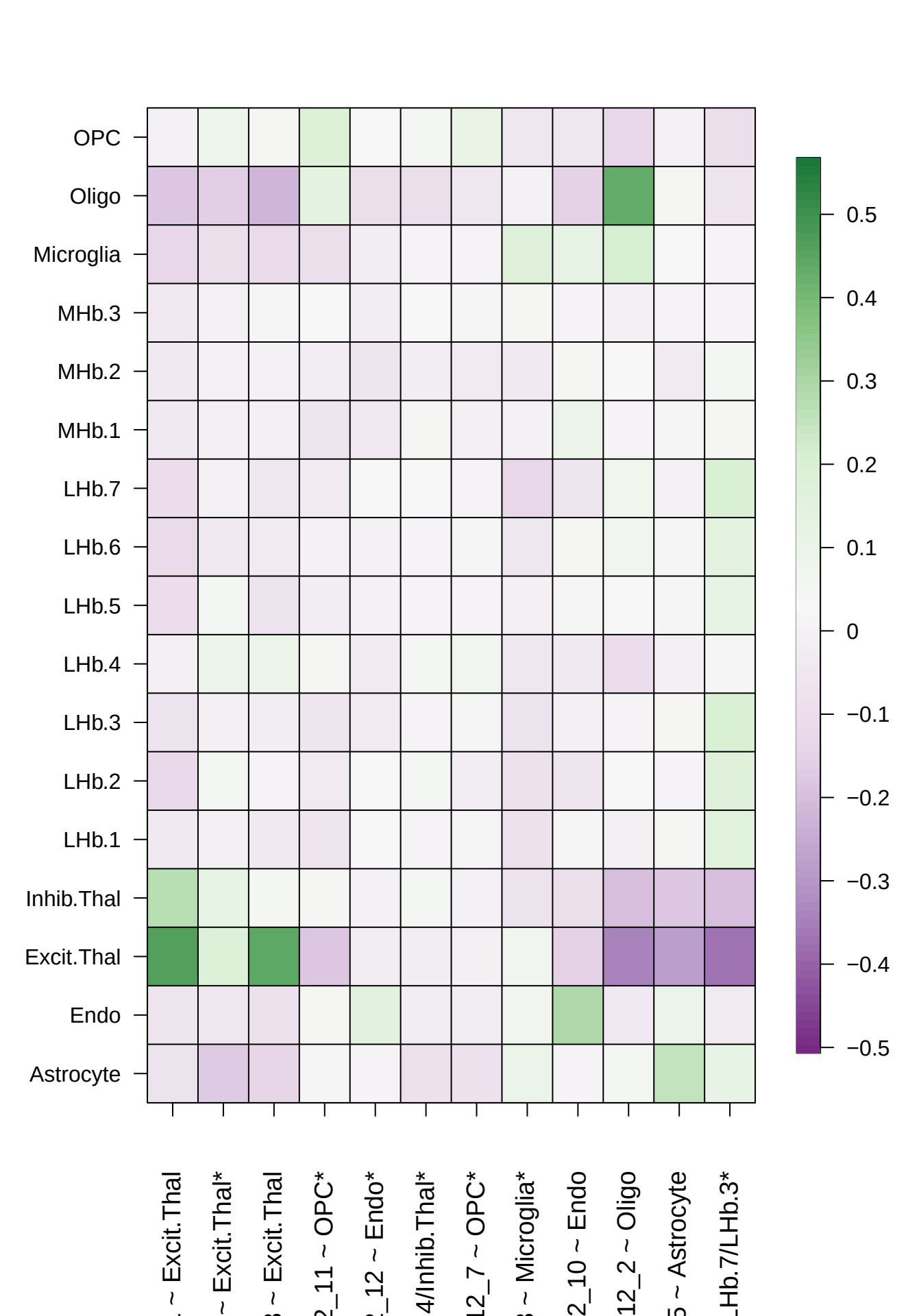

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
~ MHb.1/OPC*
_4 ~ Excit.Thal*
_1 ~ Excit.Thal
_5 ~ Excit.Thal
k07_2 ~ Oligo
k07_6 ~ Endo
k07_7 ~ OPC*
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

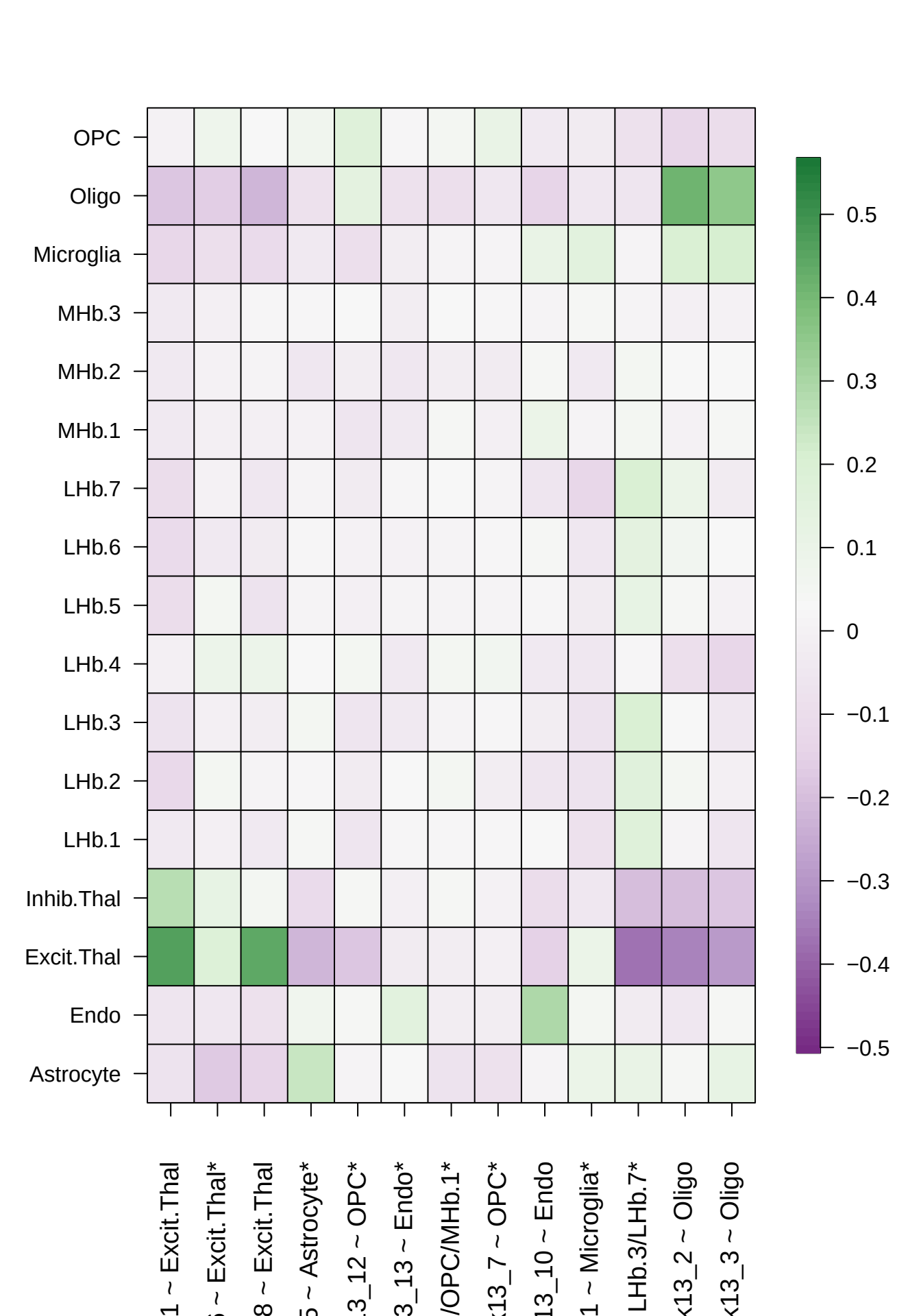

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
~ OPC/MHb.1*
_5 ~ Excit.Thal*
_1 ~ Excit.Thal
_6 ~ Excit.Thal
~ LHb.3/LHb.7*
k08_2 ~ Oligo
k08_7 ~ Endo
k08_8 ~ OPC*
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

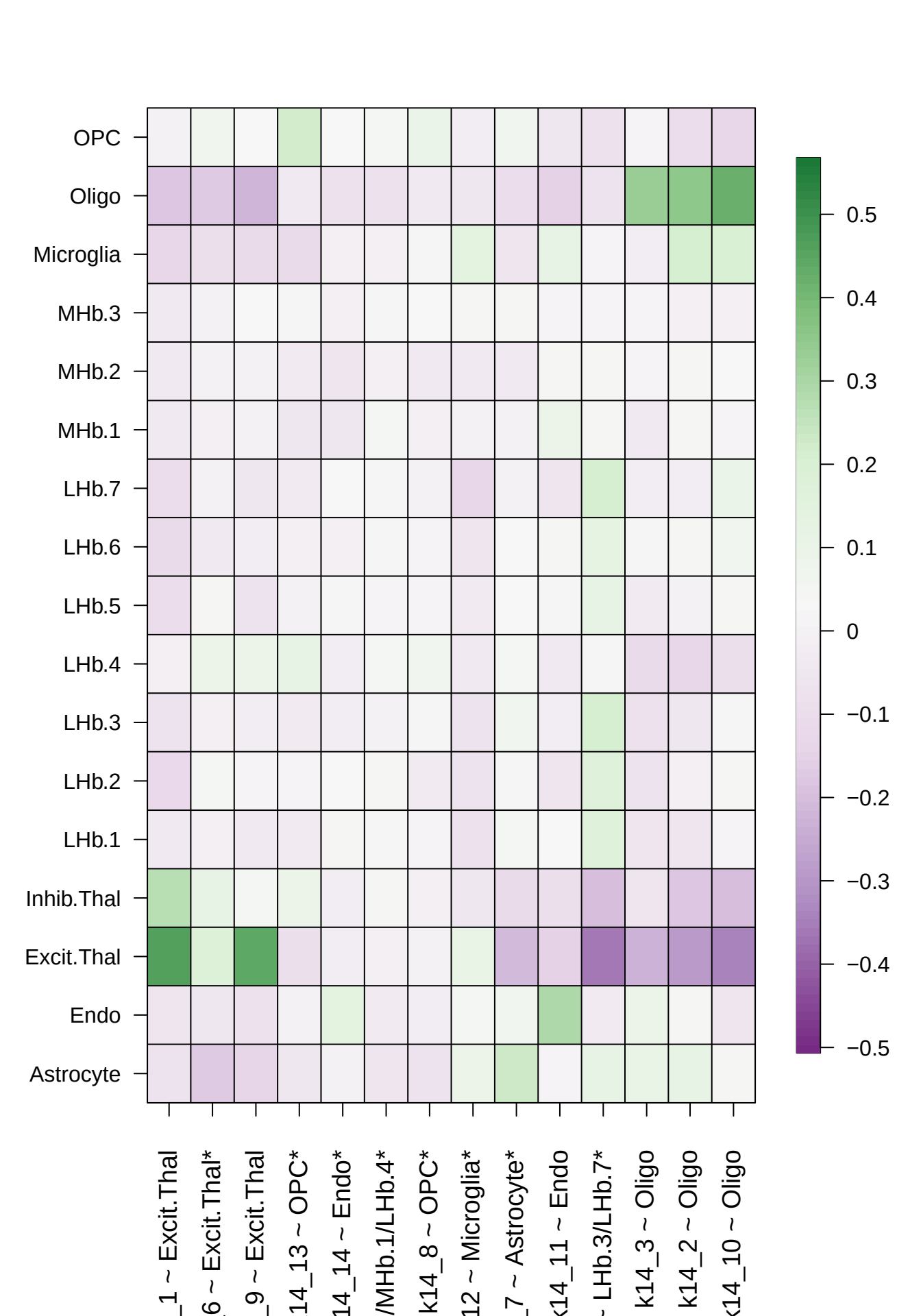

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
k09_2 ~ Oligo
~ LHb.3/LHb.7*
_3 ~ Astrocyte*
k09_8 ~ Endo
_1 ~ Excit.Thal
_7 ~ Excit.Thal
k09_9 ~ OPC*
~ OPC/MHb.1*
_6 ~ Excit.Thal*
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5


OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
k10_2 ~ Oligo
LHb.3/LHb.7*
3 ~ Astrocyte*
k10_9 ~ Endo
1 ~ Excit.Thal
8 ~ Excit.Thal
0_10 ~ OPC*
~ Excit.Thal*
ib.Thal/LHb.2*
k10_7 ~ OPC*
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

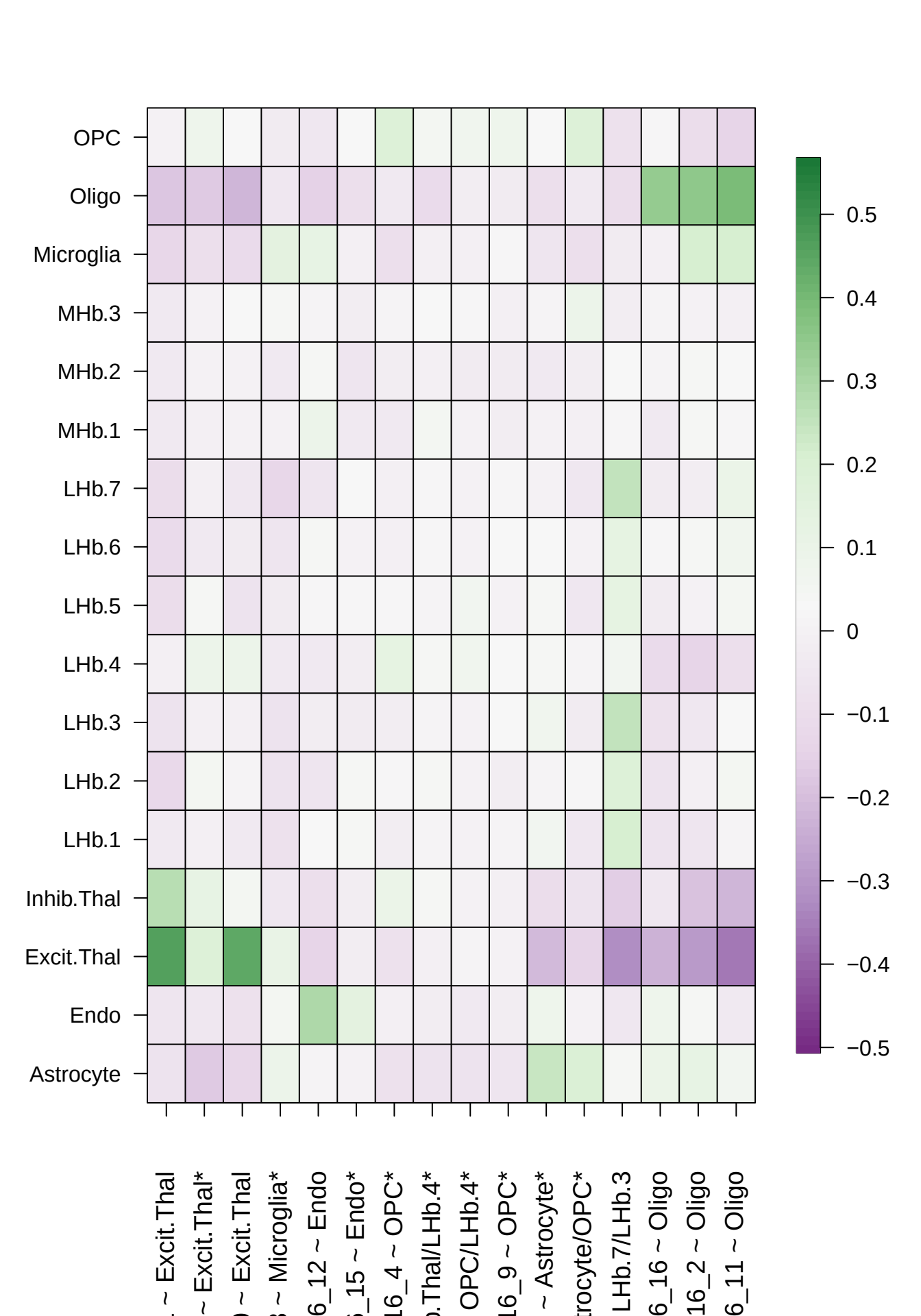
OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
~ Excit.Thal
~ Excit.Thal*
~ Excit.Thal
_10 ~ OPC*
_11 ~ Endo*
Hb.1/LHb.4*
1_7 ~ OPC*
~ Astrocyte*
.1_9 ~ Endo
.1_2 ~ Oligo
Hb.3/LHb.7*

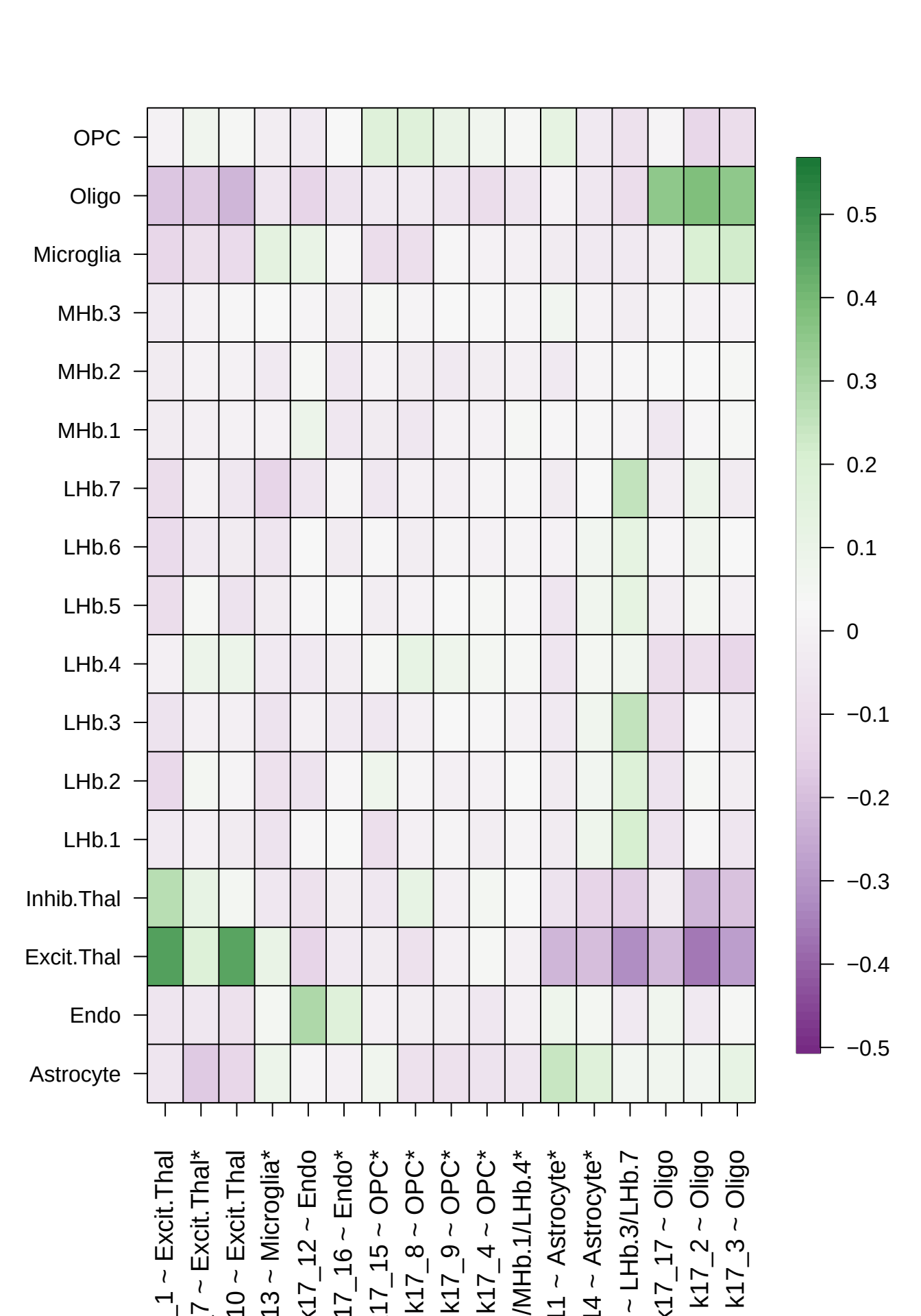

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
~ Excit.Thal
~ Excit.Thal*
~ Excit.Thal
2_11 ~ OPC*
_12 ~ Endo*
4/Inhib.Thal*
L2_7 ~ OPC*
~ Microglia*
2_10 ~ Endo
12_2 ~ Oligo
5 ~ Astrocyte
_Hb.7/LHb.3*
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

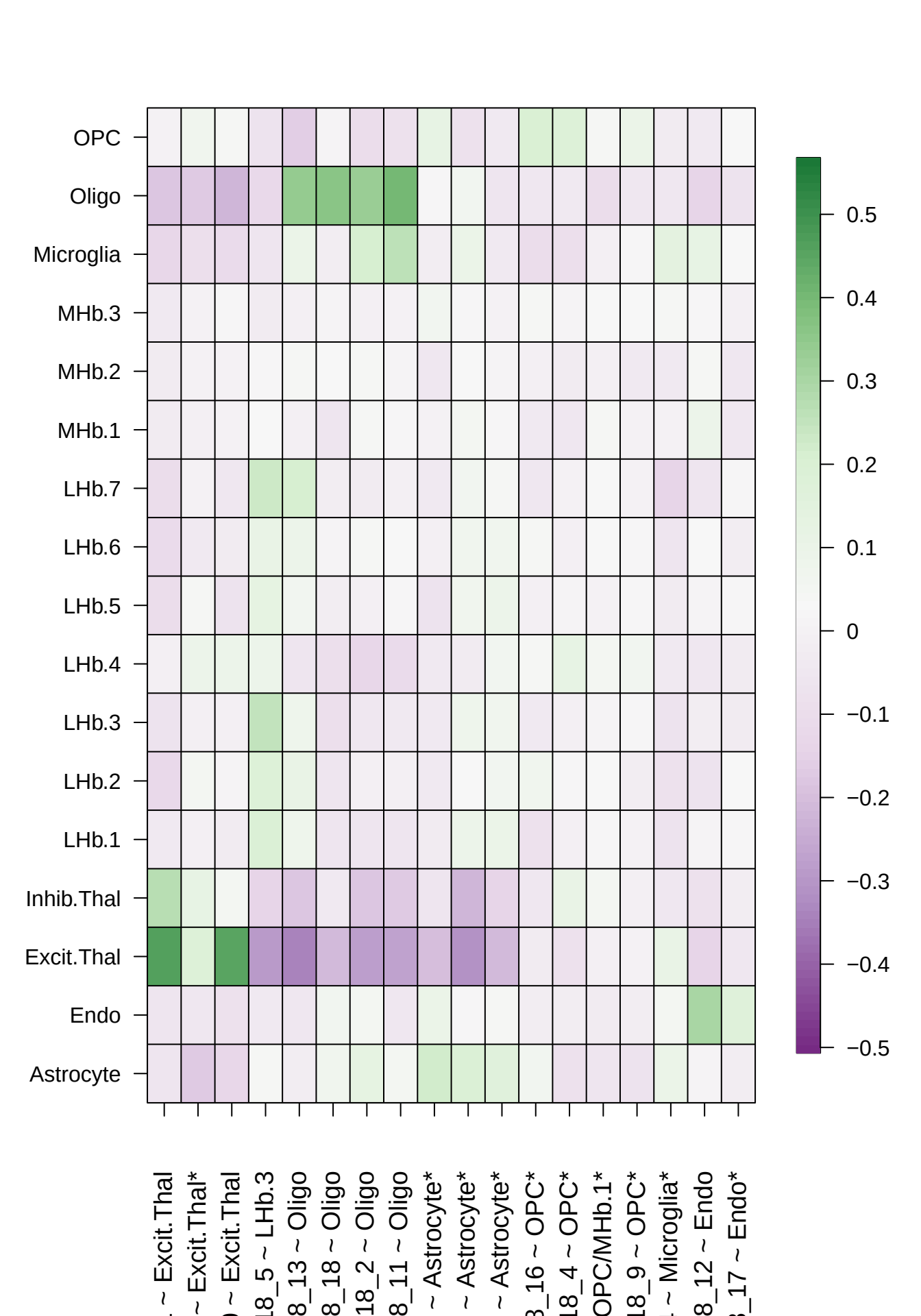

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
1 ~ Excit.Thal
~ Excit.Thal*
8 ~ Excit.Thal
5 ~ Astrocyte*
.3_12 ~ OPC*
3_13 ~ Endo*
/OPC/MHb.1*
13_7 ~ OPC*
13_10 ~ Endo
1 ~ Microglia*
LHb.3/LHb.7*
13_2 ~ Oligo
13_3 ~ Oligo
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

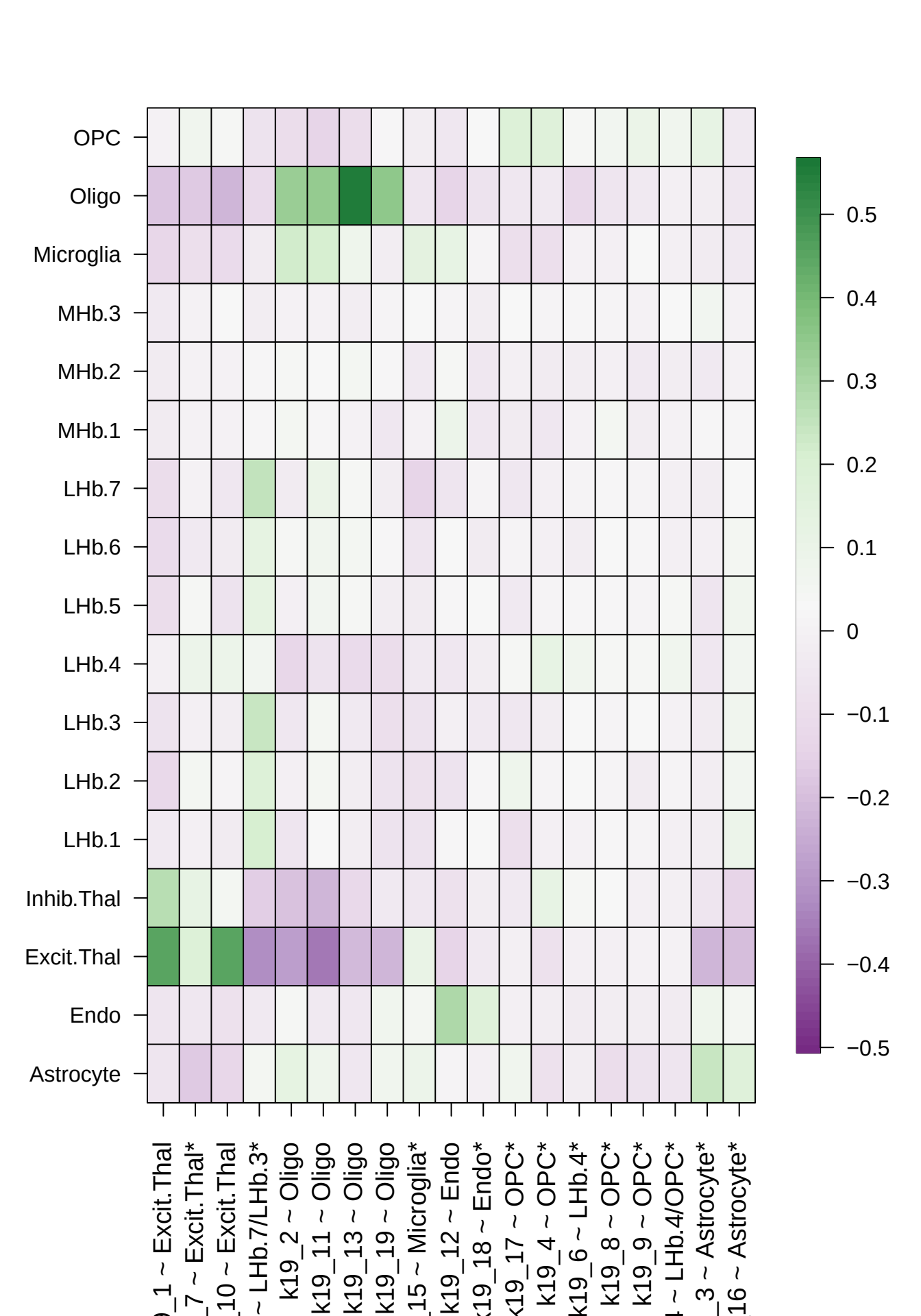

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
_1 ~ Excit.Thal
6 ~ Excit.Thal*
_9 ~ Excit.Thal
14_13 ~ OPC*
14_14 ~ Endo*
/MHb.1/LHb.4*
k14_8 ~ OPC*
12 ~ Microglia*
_7 ~ Astrocyte*
K14_11 ~ Endo
~ LHb.3/LHb.7*
k14_3 ~ Oligo
k14_2 ~ Oligo
K14_10 ~ Oligo
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

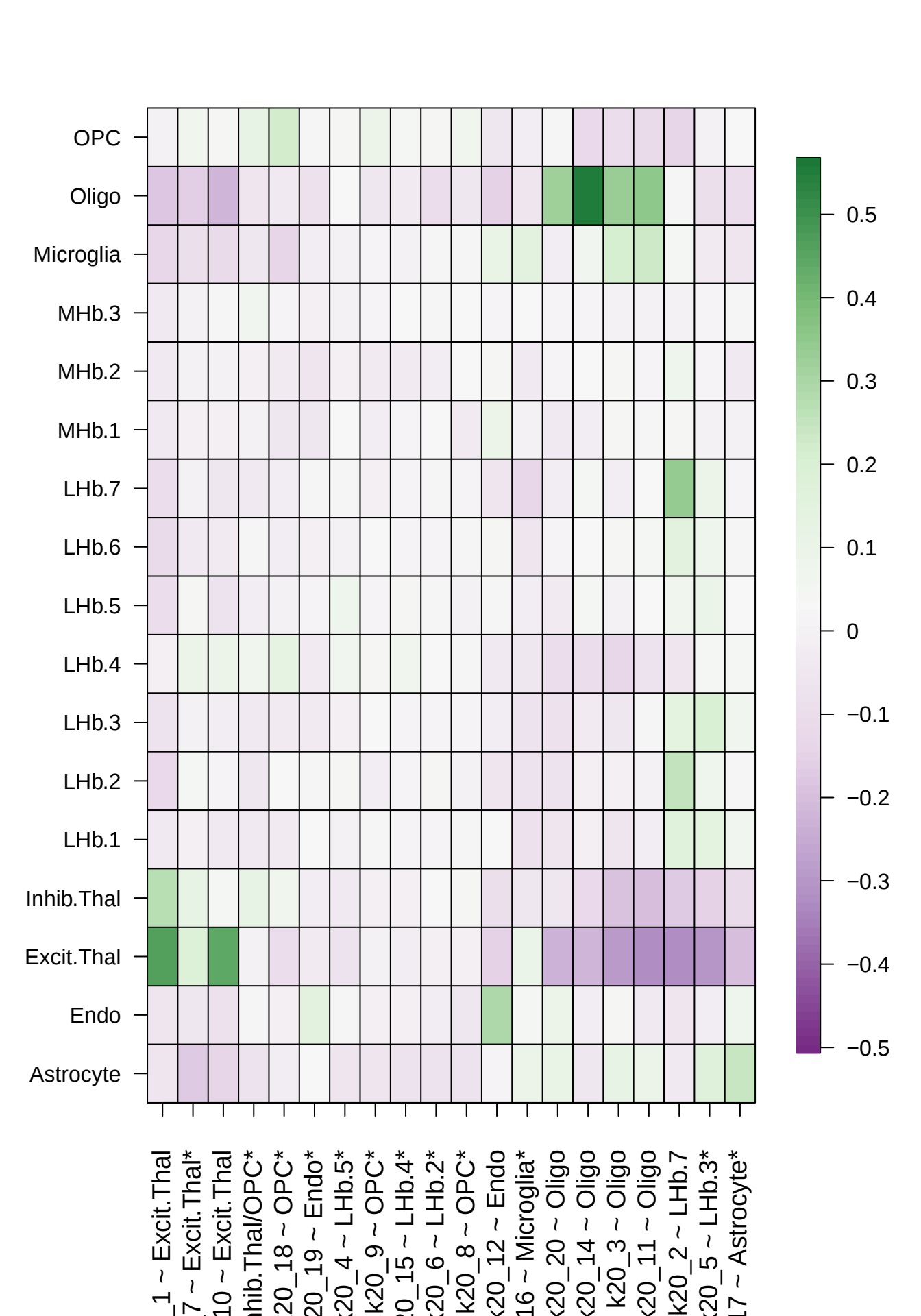
OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
k15_1 ~ Excit.Thal
k15_7 ~ Excit.Thal*
k15_10 ~ Excit.Thal
k15_5 ~ LHb.3/LHb.7
k15_8 ~ Oligo
k15_2 ~ Oligo
k15_11 ~ Oligo
k15_4 ~ OPC*
k15_6 ~ OPC*
k15_9 ~ OPC*
k15_3 ~ Astrocyte*
k15_14 ~ Astrocyte*
k15_13 ~ Microglia*
k15_12 ~ Endo
k15_15 ~ Endo*
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5

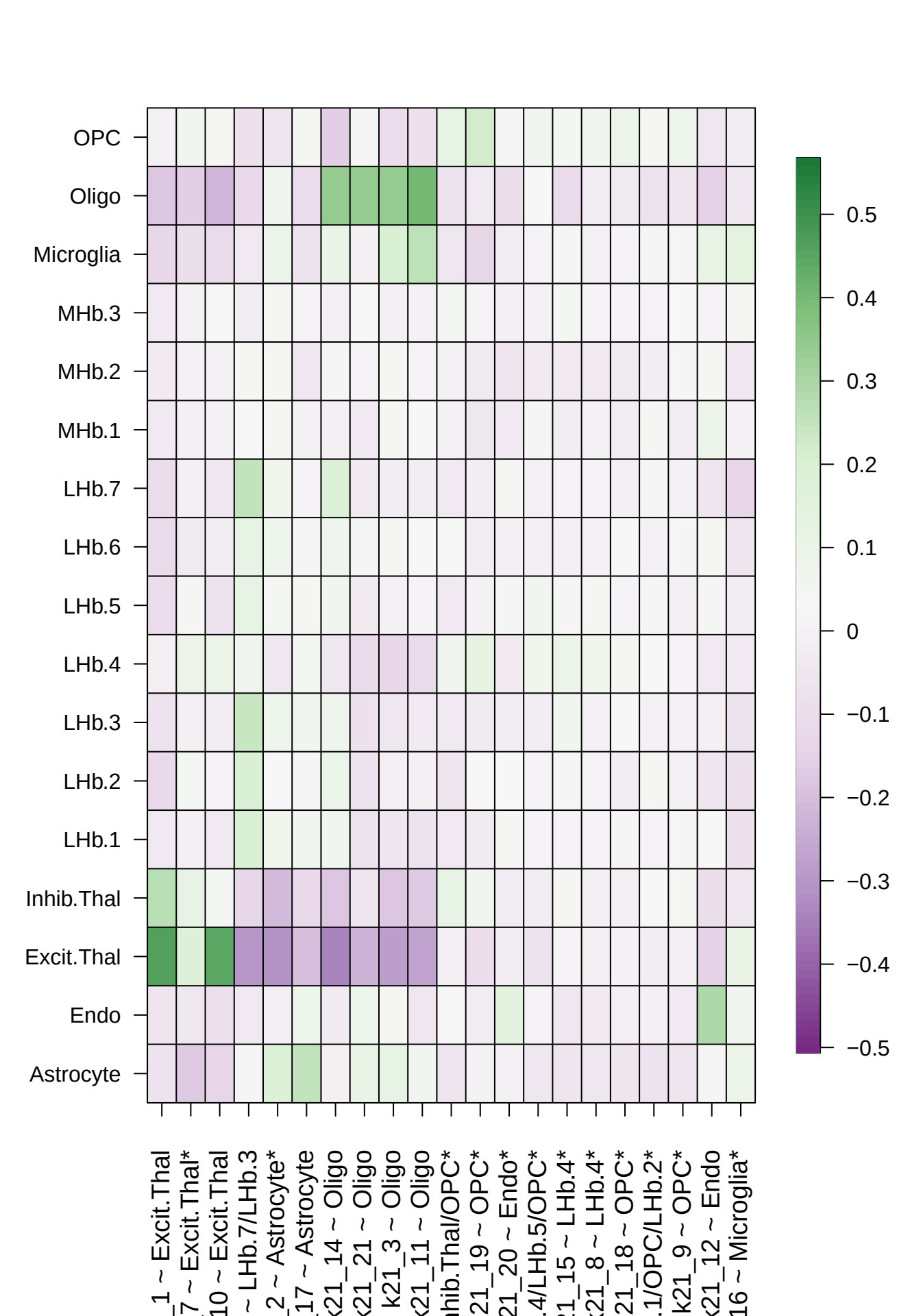

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
~ Excit.Thal
~ Excit.Thal*
~ Excit.Thal
~ Microglia*
6_12 ~ Endo
6_15 ~ Endo*
16_4 ~ OPC*
.Thal/LHb.4*
OPC/LHb.4*
16_9 ~ OPC*
~ Astrocyte*
trocyte/OPC*
LHb.7/LHb.3
6_16 ~ Oligo
16_2 ~ Oligo
6_11 ~ Oligo

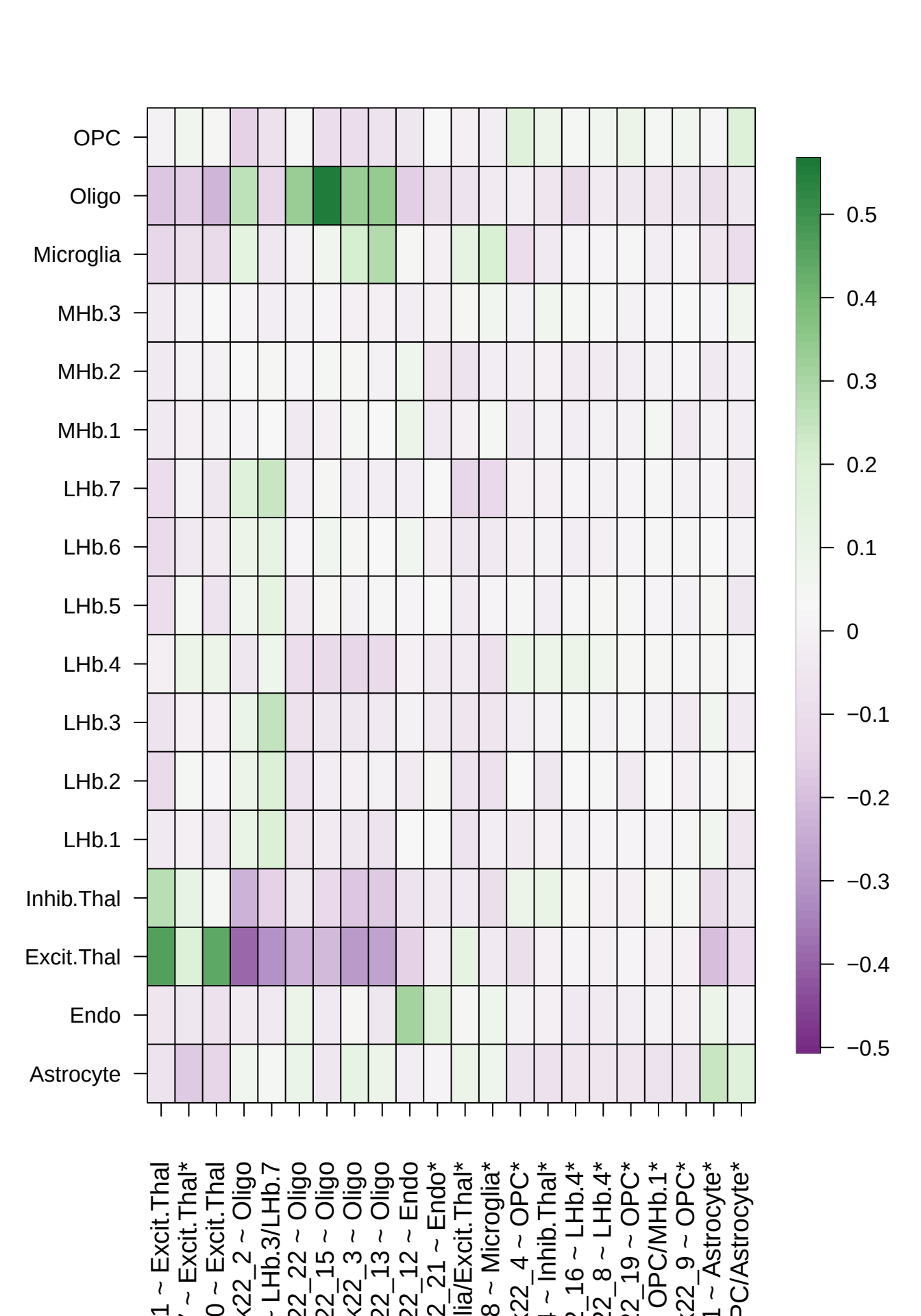

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
_1 ~ Excit.Thal
7 ~ Excit.Thal*
10 ~ Excit.Thal
13 ~ Microglia*
k17_12 ~ Endo
17_16 ~ Endo*
17_15 ~ OPC*
k17_8 ~ OPC*
k17_9 ~ OPC*
k17_4 ~ OPC*
/MHb.1/LHb.4*
L1 ~ Astrocyte*
L4 ~ Astrocyte*
~ LHb.3/LHb.7
k17_17 ~ Oligo
k17_2 ~ Oligo
k17_3 ~ Oligo

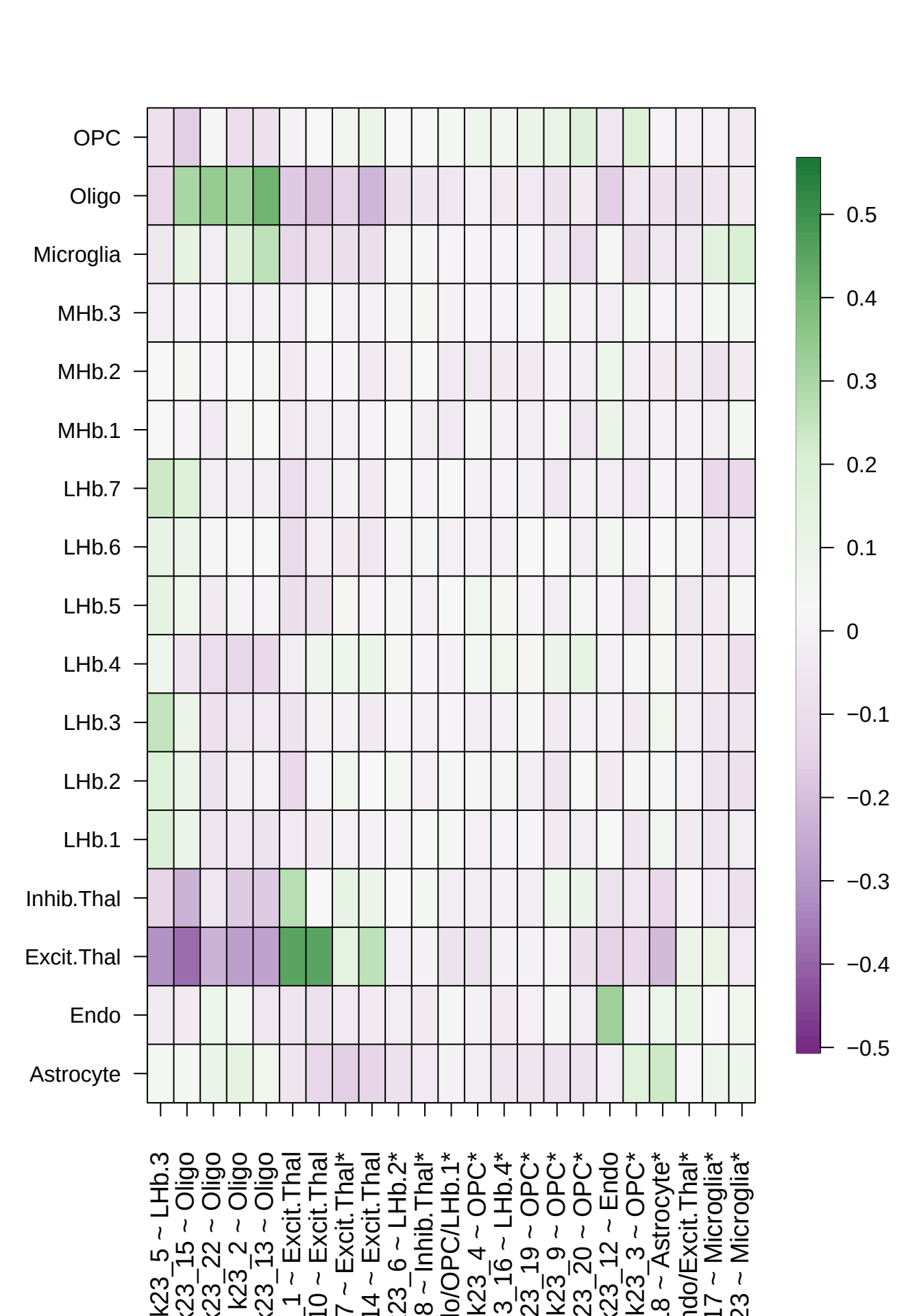

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
~ Excit.Thal
~ Excit.Thal*
~ Excit.Thal
18_5 ~ LHb.3
8_13 ~ Oligo
8_18 ~ Oligo
18_2 ~ Oligo
8_11 ~ Oligo
~ Astrocyte*
~ Astrocyte*
~ Astrocyte*
3_16 ~ OPC*
18_4 ~ OPC*
OPC/MHb.1*
18_9 ~ OPC*
~ Microglia*
8_12 ~ Endo
3_17 ~ Endo*

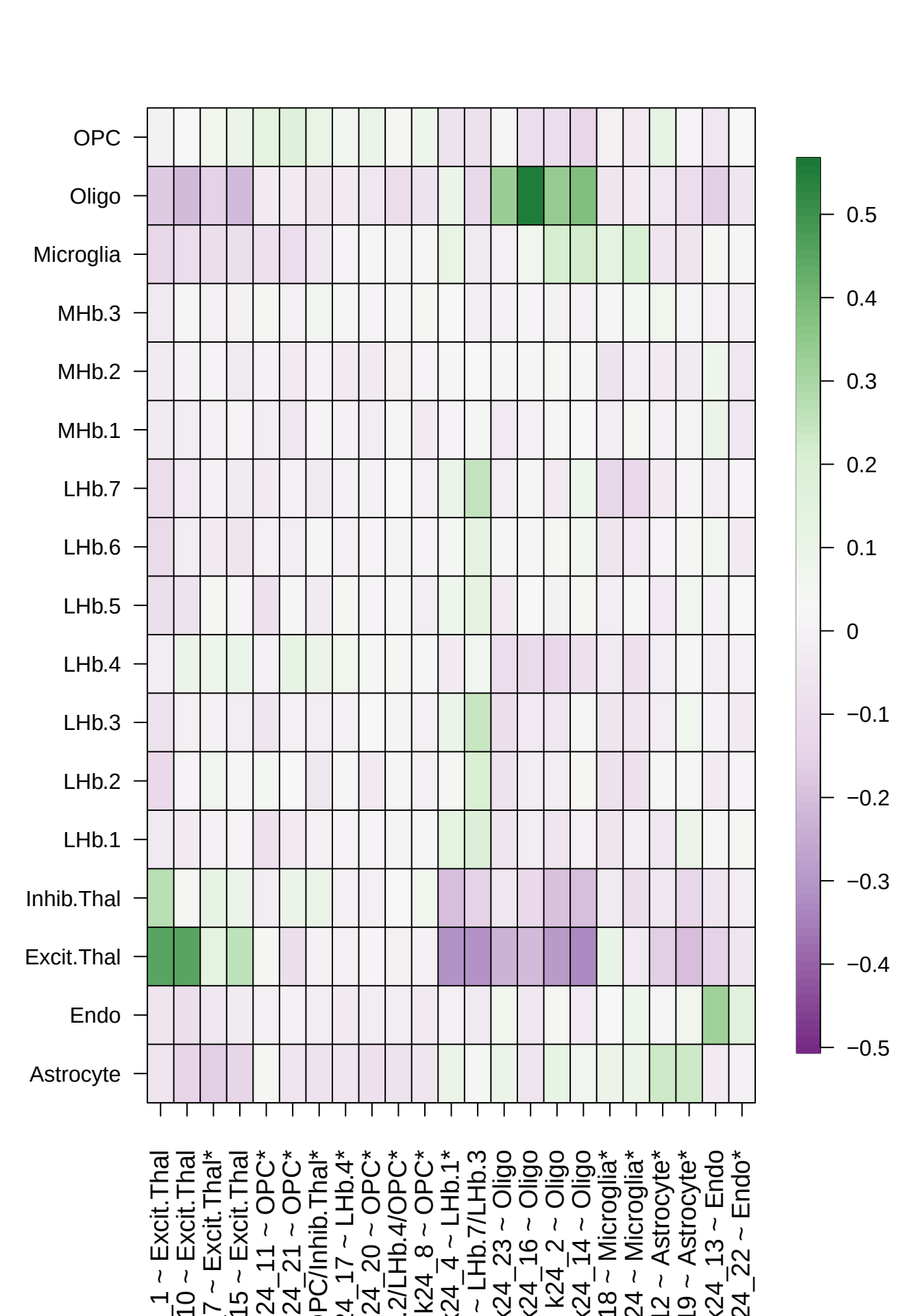

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
k19_1 ~ Excit.Thal
k19_7 ~ Excit.Thal*
k19_10 ~ Excit.Thal
~ LHb.7/LHb.3*
k19_2 ~ Oligo
k19_11 ~ Oligo
k19_13 ~ Oligo
k19_19 ~ Oligo
_15 ~ Microglia*
k19_12 ~ Endo
k19_18 ~ Endo*
k19_17 ~ OPC*
k19_4 ~ OPC*
k19_6 ~ LHb.4*
k19_8 ~ OPC*
k19_9 ~ OPC*
~ LHb.4/OPC*
_3 ~ Astrocyte*
_16 ~ Astrocyte*

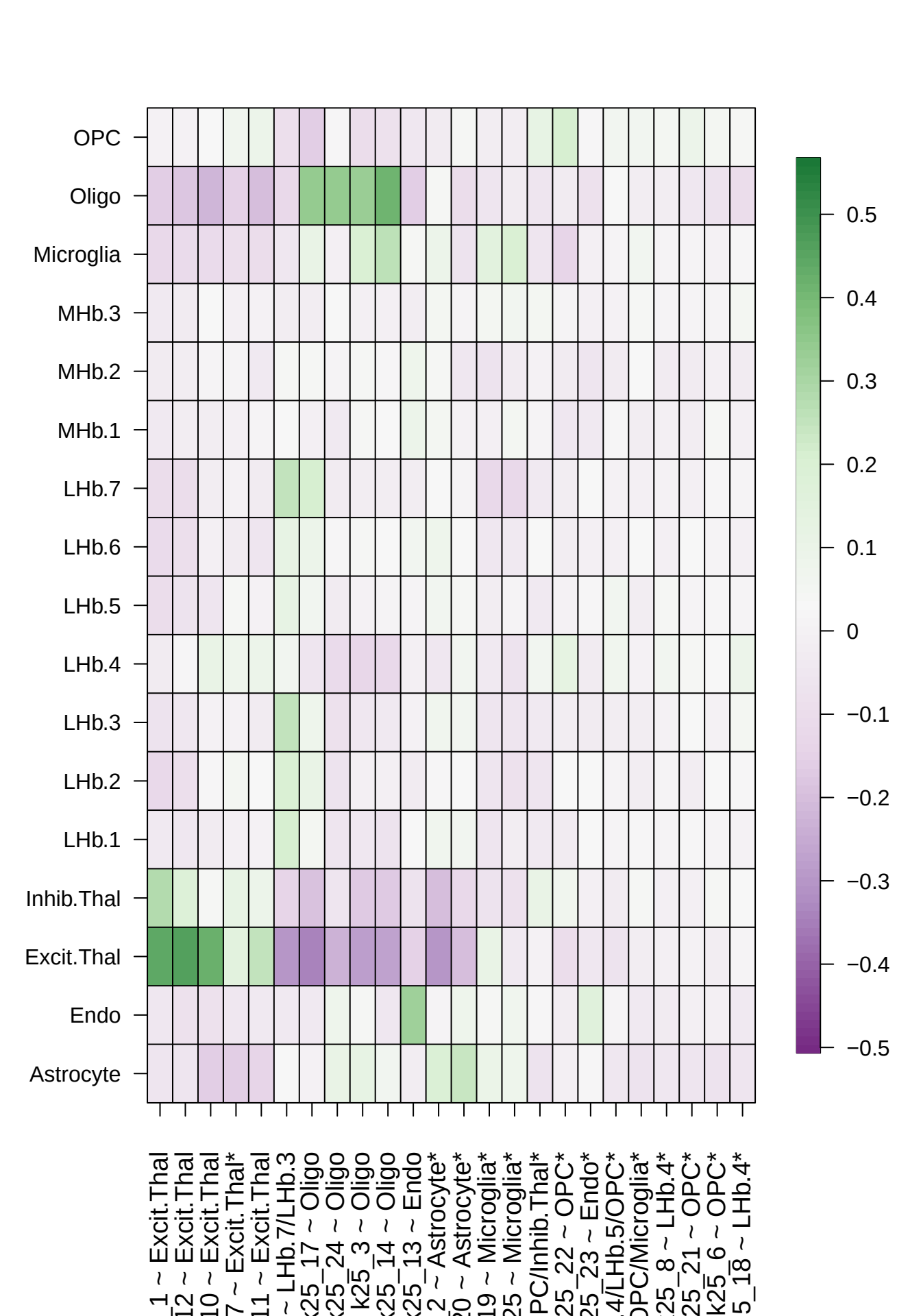

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
_1 ~ Excit.Thal
7 ~ Excit.Thal*
10 ~ Excit.Thal
nhib.Thal/OPC*
20_18 ~ OPC*
20_19 ~ Endo*
20_4 ~ LHb.5*
k20_9 ~ OPC*
20_15 ~ LHb.4*
20_6 ~ LHb.2*
k20_8 ~ OPC*
20_12 ~ Endo
16 ~ Microglia*
20_20 ~ Oligo
20_14 ~ Oligo
k20_3 ~ Oligo
20_11 ~ Oligo
k20_2 ~ LHb.7
20_5 ~ LHb.3*
L7 ~ Astrocyte*

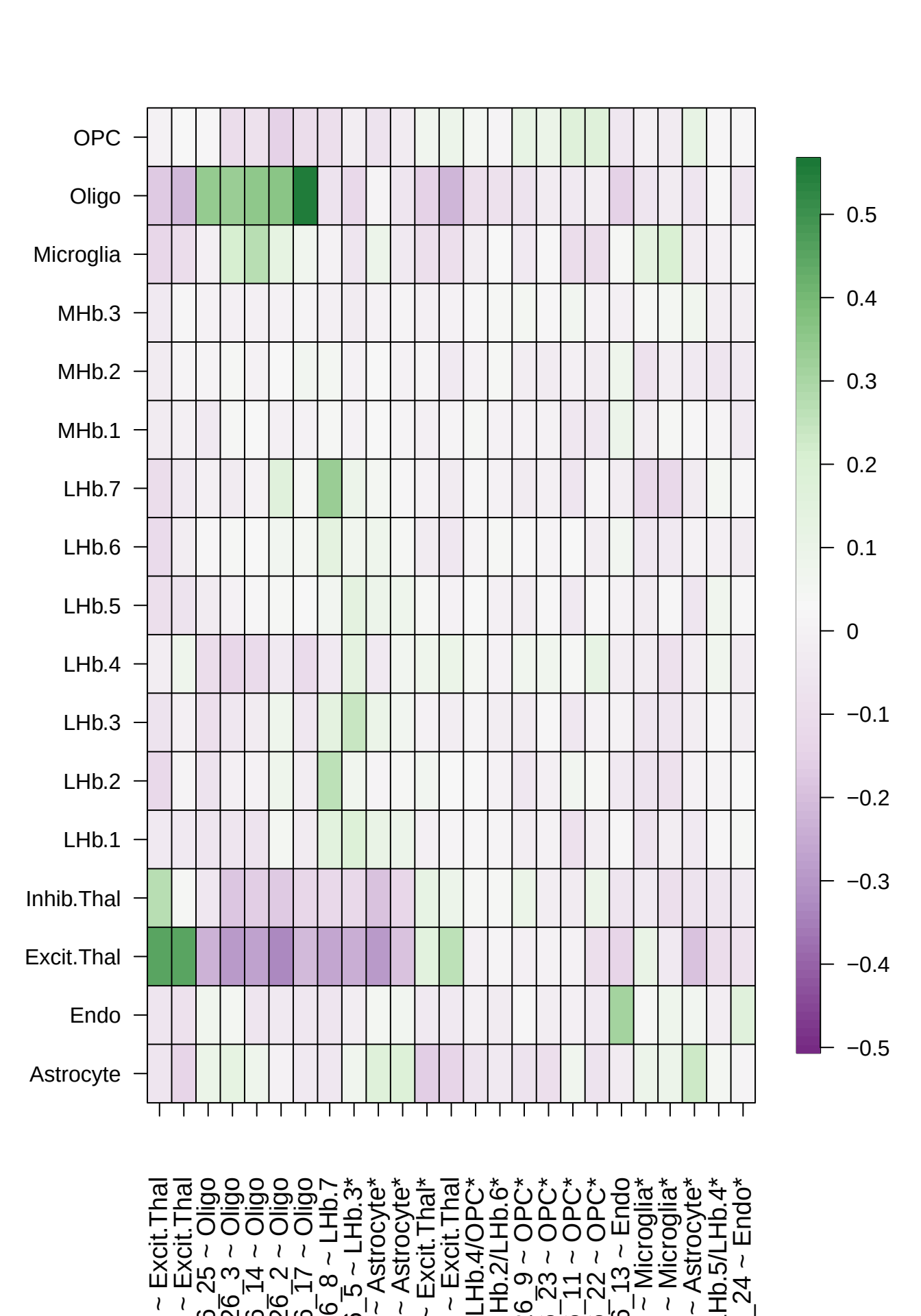

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
_1 ~ Excit.Thal
_7 ~ Excit.Thal*
_10 ~ Excit.Thal
~ LHb.7/LHb.3
_2 ~ Astrocyte*
_17 ~ Astrocyte
K21_14 ~ Oligo
K21_21 ~ Oligo
k21_3 ~ Oligo
K21_11 ~ Oligo
hib.Thal/OPC*
21_19 ~ OPC*
21_20 ~ Endo*
.4/LHb.5/OPC*
21_15 ~ LHb.4*
21_8 ~ LHb.4*
21_18 ~ OPC*
.1/OPC/LHb.2*
k21_9 ~ OPC*
K21_12 ~ Endo
16 ~ Microglia*

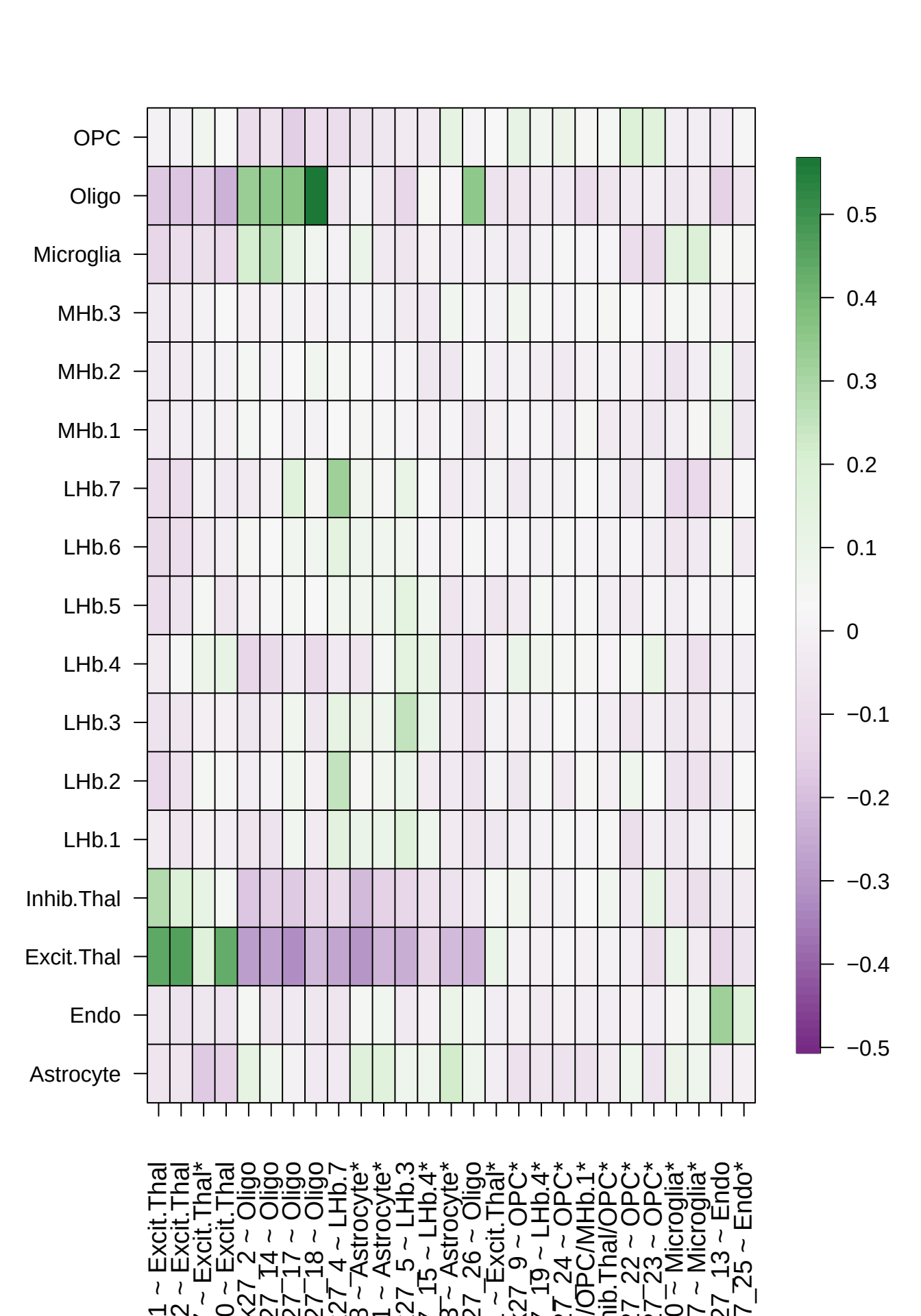

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
Excit.Thal*
Excit.Thal*
Excit.Thal
Oligo
LHb.3/LHb.7
Oligo
Oligo
Oligo
Endo
Endo*
Excit.Thal*
Microglia*
OPC*
Inhib.Thal*
LHb.4*
LHb.4*
OPC*
OPC/MHb.1*
OPC*
Astrocyte*
OPC/Astrocyte*

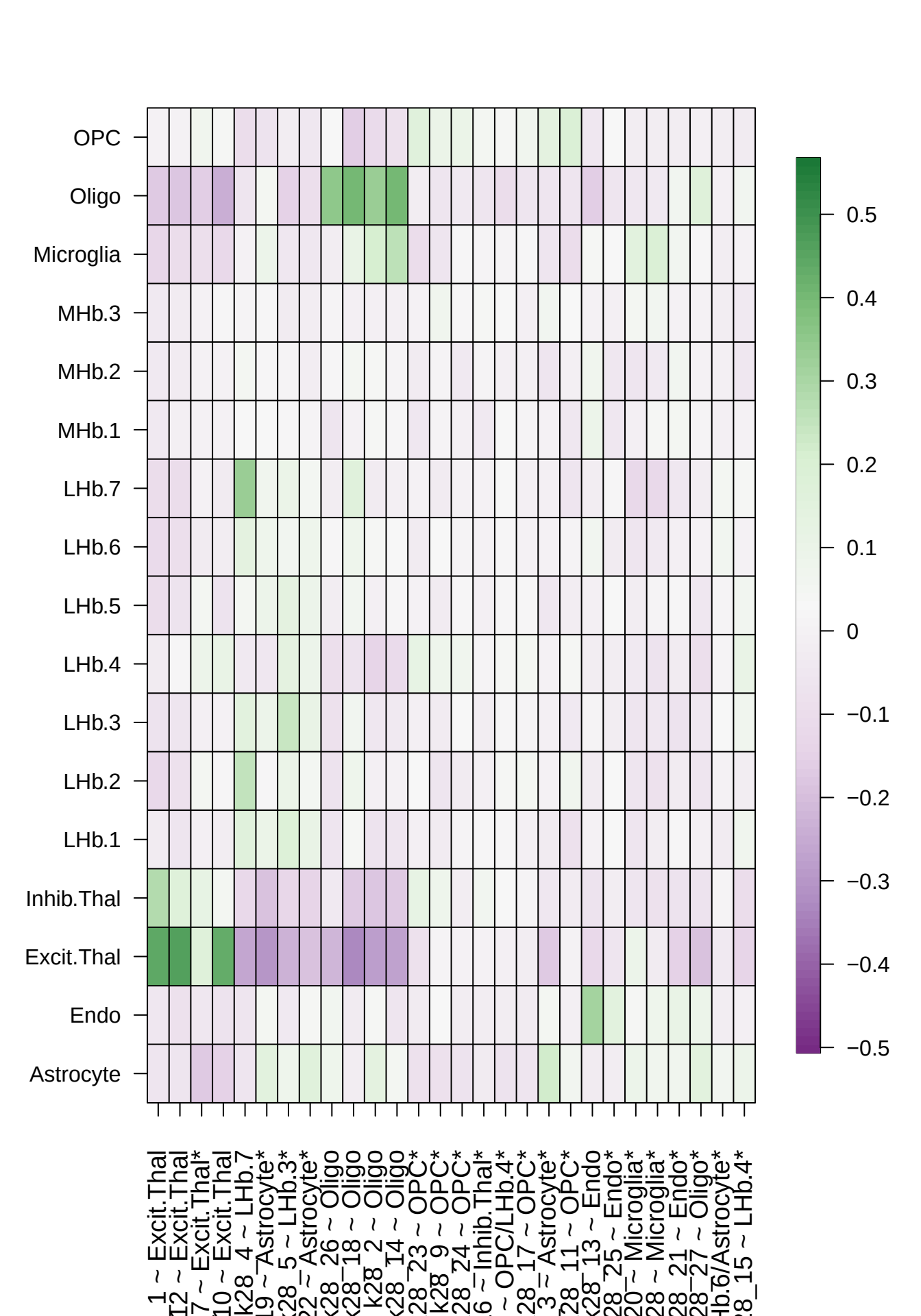

OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
k23_5 ~ LHb.3
k23_15 ~ Oligo
k23_22 ~ Oligo
k23_2 ~ Oligo
k23_13 ~ Oligo
_1 ~ Excit.Thal
10 ~ Excit.Thal
7 ~ Excit.Thal*
14 ~ Excit.Thal
23_6 ~ LHb.2*
8 ~ Inhib.Thal*
lo/OPC/LHb.1*
k23_4 ~ OPC*
3_16 ~ LHb.4*
23_19 ~ OPC*
k23_9 ~ OPC*
23_20 ~ OPC*
k23_12 ~ Endo
k23_3 ~ OPC*
8 ~ Astrocyte*
ndo/Excit.Thal*
17 ~ Microglia*
23 ~ Microglia*


OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
_1 ~ Excit.Thal
_10 ~ Excit.Thal
_7 ~ Excit.Thal*
_15 ~ Excit.Thal
24_11 ~ OPC*
24_21 ~ OPC*
PC/Inhib.Thal*
4_17 ~ LHb.4*
24_20 ~ OPC*
.2/LHb.4/OPC*
k24_8 ~ OPC*
24_4 ~ LHb.1*
~ LHb.7/LHb.3
k24_23 ~ Oligo
k24_16 ~ Oligo
k24_2 ~ Oligo
k24_14 ~ Oligo
18 ~ Microglia*
24 ~ Microglia*
L2 ~ Astrocyte*
L9 ~ Astrocyte*
k24_13 ~ Endo
24_22 ~ Endo*


OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
_1 ~ Excit.Thal
12 ~ Excit.Thal
10 ~ Excit.Thal
7 ~ Excit.Thal*
11 ~ Excit.Thal
~ LHb.7/LHb.3
25_17 ~ Oligo
25_24 ~ Oligo
k25_3 ~ Oligo
25_14 ~ Oligo
25_13 ~ Endo
2 ~ Astrocyte*
0 ~ Astrocyte*
19 ~ Microglia*
25 ~ Microglia*
PC/Inhib.Thal*
25_22 ~ OPC*
25_23 ~ Endo*
4/LHb.5/OPC*
OPC/Microglia*
25_8 ~ LHb.4*
25_21 ~ OPC*
k25_6 ~ OPC*
5_18 ~ LHb.4*


OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
~ Excit.Thal
~ Excit.Thal
_25 ~ Oligo
_3 ~ Oligo
_14 ~ Oligo
_2 ~ Oligo
_17 ~ Oligo
_8 ~ LHb.7
_5 ~ LHb.3*
~ Astrocyte*
~ Astrocyte*
~ Excit.Thal*
~ Excit.Thal
LHb.4/OPC*
Hb.2/LHb.6*
_9 ~ OPC*
_23 ~ OPC*
_11 ~ OPC*
_22 ~ OPC*
_13 ~ Endo
~ Microglia*
~ Microglia*
~ Astrocyte*
Hb.5/LHb.4*
_24 ~ Endo*


OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
Excit.Thal
Excit.Thal
Excit.Thal*
Excit.Thal
Oligo
Oligo
Oligo
LHb.7
Astrocyte*
Astrocyte*
LHb.3
LHb.4*
Astrocyte*
Oligo
Excit.Thal*
OPC*
LHb.4*
OPC*
OPC/MHb.1*
Inhib.Thal/OPC*
OPC*
OPC*
OPC*
Microglia*
Microglia*
Endo
Endo*


OPC
Oligo
Microglia
MHb.3
MHb.2
MHb.1
LHb.7
LHb.6
LHb.5
LHb.4
LHb.3
LHb.2
LHb.1
Inhib.Thal
Excit.Thal
Endo
Astrocyte
0.5
0.4
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
1 ~ Excit.Thal
I2 ~ Excit.Thal
7 ~ Excit.Thal*
10 ~ Excit.Thal
k28_4 ~ LHb.7
L9 ~ Astrocyte*
28_5 ~ LHb.3*
2 ~ Astrocyte*
28_26 ~ Oligo
28_18 ~ Oligo
k28_2 ~ Oligo
28_14 ~ Oligo
28_23 ~ OPC*
k28_9 ~ OPC*
28_24 ~ OPC*
6 ~ Inhib.Thal*
~ OPC/LHb.4*
28_17 ~ OPC*
3 ~ Astrocyte*
28_11 ~ OPC*
28_13 ~ Endo
28_25 ~ Endo*
20 ~ Microglia*
28 ~ Microglia*
28_21 ~ Endo*
28_27 ~ Oligo*
b.6/Astrocyte*
8_15 ~ LHb.4*